NEXT-PHASE ROADMAP

Next-Phase Architecture & Expansion

Enterprise Inventory Management System

A feature expansion roadmap covering security and role-based access control, four advanced enterprise capabilities, and an event-driven n8n webhook automation pipeline.

IDENTITY
ASP.NET Core Identity

ACCESS MODEL
5 Roles · RBAC

EXPANSIONS
4 Enterprise Features

AUTOMATION
n8n Webhooks

ROADMAP SCOPE

Security & RBAC | Barcode Engine | Returns & Scrap | Bulk Excel / CSV Import | Multi-Warehouse | n8n Automation

OVERVIEW

Table of Contents

Security, advanced features, and webhook automation.

SECTION 1 · SECURITY, IDENTITY & RBAC

1.1 **Security Baseline & Authentication** — 03
Identity provider, cookie & JWT auth, password policy, claims

1.2 **User Roles & Access Control Matrix** — 04
Five roles with scoped permissions and boundaries

SECTION 2 · ADVANCED ENTERPRISE FEATURE EXPANSIONS

2.1 **Barcode Engine & Label Printing** — 05
Keyboard-wedge POS scanning, Code 128 / QR label sheets

2.2 **Inventory Returns & Scrap Adjustments** — 05
Refund workflow, atomic reversal, damage & expiry journal

2.3 **Bulk Data Import via Excel / CSV** — 06
ClosedXML / CsvHelper ingestion with pre-flight validation

2.4 **Multi-Warehouse & Stock Transfers** — 06
Warehouse entity and inter-warehouse transfer orders (STO)

SECTION 3 · EXTERNAL WEBHOOK AUTOMATION (N8N)

3.1 **Event-Driven Notification Pipeline** — 07
ASP.NET Core → HMAC-signed webhook → n8n → Telegram / WhatsApp

3.2 **Integrated Automation Workflows** — 08
Low-stock dispatch, daily executive digest, critical warnings

SECTION 01 · SECURITY

Security & Identity Baseline

Technical stack and security foundation required before any new feature is rolled out.

1

IDENTITY PROVIDER

ASP.NET Core Identity integrated with **Entity Framework Core**.

2

AUTHENTICATION MECHANISMS

- **Cookie Authentication**: secure HTTP-only cookies for web portal sessions.
- **JWT Bearer Tokens**: signed tokens (HMAC-SHA256 or RSA) to support external API consumers, mobile handhelds, and decoupled endpoints.

3

PASSWORD POLICIES

Enforced entropy standards using **PBKDF2** hashing with dynamic salting.

4

AUTHORIZATION ENFORCEMENT

Policy-based requirements mapped to fine-grained claims, for example:

`Permissions.Inventory.Edit` `Permissions.Sales.Execute`

AUTH SCHEMES	HASHING	AUTHORIZATION
Cookie + JWT	PBKDF2 + Salt	Policy + Claims

NEXT · SECTION 01
User Roles & Access Control Matrix →

SECTION 01 · SECURITY

User Roles & Access Matrix

Five roles, each with a clearly scoped set of permissions and boundaries.

1	**Super Admin / Business Owner** Executive Management	Full unrestricted system control. Manages application users, role assignments, financial audit trails, tenant settings, and irreversible records.	FULL ACCESS
2	**Warehouse / Inventory Manager** Logistics Lead	Comprehensive authority over Products, Categories, Suppliers, Supplier Mappings, and Purchase Orders (Stock-In). Restricted from POS customer transactions.	MANAGE
3	**Cashier / POS Operator** Storefront Staff	Dedicated POS checkout, rapid item entry, stock-out invoices, and receipt printing. No access to purchase costs, profit margins, or catalog modification.	POS ONLY
4	**Procurement Specialist** Purchasing Officer	Supplier directories, purchase order drafting, contract pricing evaluation, and lead-time monitoring. Excluded from retail sales operations.	PURCHASING
5	**Auditor / Financial Viewer** Accounting & Compliance	Read-only visibility across Financial Reports, Stock Valuation, Sales Performance, and historical invoice verification. Cannot alter database records.	READ-ONLY

ACCESS BOUNDARIES

- Cashiers never see vendor purchase costs or profit margins.
- Warehouse and Procurement roles are kept out of retail sales operations.
- The Auditor role is strictly read-only across all financial data.

NEXT · SECTION 02
Barcode Engine, Returns & Scrap →

SECTION 02 · FEATURES

Advanced Feature Expansions

Barcode tooling and inventory correction workflows.

2.1 Barcode Engine & Label Printing

BARCODE & LABELS

HARDWARE-INTEGRATED POS SCANNING

- Built-in keyboard wedge input listener on the POS terminal.
- Captures scanner input and extracts the product **SKU**.
- Appends the item to the transaction grid with zero keyboard intervention.

AUTOMATED LABEL & BARCODE GENERATION

- Client and server-side generation supporting **Code 128** and **QR Code**.
- Export layout engine producing printable adhesive label sheets for warehouse shelves and product packaging.

2.2 Inventory Returns & Scrap Adjustments

RETURNS & SCRAP

SALES RETURN & REFUND WORKFLOW

- Structured refund pipeline tied to the original sales invoice number (**SaleID**).
- Automatic inventory reversal restoring returned units to stock under atomic database transactions.
- Mandatory audit notes tracking reasons (e.g., Defective Item, Customer Exchange).

DAMAGE & EXPIRY SCRAP JOURNAL

- Non-sale write-off journal to decommission damaged, expired, or obsolete goods.
- Reconciles physical stock with ledger counts without distorting sales revenue figures.

NEXT · SECTION 02
Bulk Import & Multi-Warehouse →

Advanced Feature Expansions

Mass data ingestion and multi-branch inventory distribution.

2.3 Bulk Data Import via Excel / CSV

BULK IMPORT

HIGH-THROUGHPUT BATCH PROCESSING

- Ingestion service using **ClosedXML** / **CsvHelper** for mass seeding of products, categories, and supplier catalogs.
- Wrapped in isolated database transactions for an all-or-nothing commit model.

PRE-INGESTION VALIDATION PIPELINE

- Real-time pre-flight checks catching duplicate SKUs, broken foreign key references, and invalid numerical bounds before database execution.

2.4 Multi-Warehouse & Stock Transfers

MULTI-WAREHOUSE

MULTI-BRANCH DATA STRUCTURE

- New relational **Warehouse** entity distributing stock counts per facility (e.g., Central Depot, Downtown Branch, Regional Facility).

INTER-WAREHOUSE TRANSFER ORDERS (STO)

- Formally tracked movement lifecycle with dual-state status tracking:

Draft → **Pending Approval** → **In-Transit** → **Received**

- Full custody tracking ensures inventory is never lost between physical locations.

NEXT · SECTION 03
Event-Driven Notification Pipeline →

Webhook Automation Architecture

Event-driven notifications routed through n8n to Telegram and WhatsApp.



PIPELINE NOTES

- The application emits domain events and background-service triggers; it never talks to messaging platforms directly.
- Every payload is signed with an HMAC secret so n8n can verify its origin.

NEXT · SECTION 03
Integrated Automation Workflows →

SECTION 03 · AUTOMATION

Integrated Automation Workflows

Three automated workflows dispatched through n8n.

TRIGGER · INSTANT

Instant Low-Stock Dispatches

Triggered automatically whenever a transaction results in the low-stock condition:

`StockQuantity ≤ LowStockThreshold`

Sends an immediate, actionable alert containing:

- Product Name
- SKU
- Current Count
- Primary Supplier phone link

TRIGGER · SCHEDULED 23:59

Daily Executive Digest

Scheduled background execution via **IHostedService** or **Quartz.NET**, dispatching an end-of-day summary at **23:59** to managerial channels:

- Total daily revenue
- Completed order volume
- Top-selling items

TRIGGER · EXCEPTIONS

Critical Security & Fault Warnings

Captures severe server exceptions and unauthorized privilege escalation attempts, routing immediate alerts to administrative staff without manual log inspection.

- Server exceptions
- Privilege escalation attempts

END OF ROADMAP

Back to table of contents →